

Cardiovascular Risk: Lifestyle, Blood Pressure & Global Hypertension Context

Final Exam Data Science Project - Technical Report

Problem formulation

This project investigates whether lifestyle and basic clinical indicators can explain and predict cardiovascular disease risk. The real-world problem is translated into a binary classification task: predicting whether cardiovascular disease is present.

Data sources

Source 1 is the Kaggle Cardiovascular Disease Dataset with patient-level clinical and lifestyle variables. Source 2 is the WHO / Our World in Data hypertension prevalence dataset, which provides country-level public-health context.

Hypotheses

H1: BMI and blood pressure increase risk. H2: Physical activity reduces risk. H3: Multivariable models outperform a single-factor baseline. H4: Patient-level blood-pressure signals align with global hypertension patterns.

Mathematical background

The project uses Pearson correlation, logistic regression, random forest classification, confusion matrices, ROC curves and ROC-AUC. Logistic regression estimates $P(Y=1|X)$ using the sigmoid function.

Methods

The workflow includes data validation, clinical outlier filtering, feature engineering, exploratory analysis, model training, cross-validation, model comparison and interpretation.

Evaluation

Models are evaluated with accuracy, precision, recall, F1-score and ROC-AUC. A single-factor hypertension model is used as a baseline to test whether a multivariable model adds value.

Limitations

The data is observational, so the analysis does not prove causality. Patient-level and country-level data have different granularities and are used together for contextual interpretation, not direct row-level merging.

Conclusion

The project is designed to satisfy the exam requirements while also functioning as a portfolio-ready data science project. Its strength is the complete workflow: problem formulation, math, code, analysis,

interpretation and documentation.

Repository checklist

Requirement	Included
Jupyter notebook	Yes
Two independent data sources	Yes
Math explanation	Yes
Python code	Yes
GitHub README	Yes
10+ meaningful commits plan	Yes
Presentation	Yes